

Computational supplement scaffold for Residual cohomology and causal transport in free pure-Weyl gravity

GPT-5.6.sol*

July 15, 2026

Abstract

This tracked scaffold records the computational material assigned to the supplement in the major-revision split of `conformal-residual-cohomology.tex`. It is not an independent theorem paper yet. The archival monolith remains authoritative until the residual paper, causal paper and this supplement each have standalone claim ledgers and dependency manifests.

1 Scope and publication split

The section-level extraction map and submission gates are maintained in `../notes/conformal-paper-split-roadmap.md`. The supplement will serve two articles:

1. the residual cohomology and pairing calculation;
2. the support-local causal curvature/tractor bridge which transports that result to covariant cohomology.

It will contain implementation receipts, not a third competing theorem statement.

2 Independent central residual-rank check

The command

```
python3 symbolic/verify_conformal_residual_rank53_independent.py
```

uses only the Python standard library. It constructs the integral spin-two $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ action on

$$\mathrm{Sym}^2((2,0) \oplus (0,2)), \quad 55 = 15 + 25 + 15,$$

and obtains exact block ranks

$$53 = 14 + 25 + 14.$$

The two exact kernel vectors are the chiral binary-quartic singlets

$$e_0 e_4 - 4e_1 e_3 + 3e_2^2.$$

*OpenAI model. Contributed the research, derivations, verification code, and manuscript. Commissioned and directed by Asger Alstrup Palm (asger@area9.dk), the non-technical orchestrator and corresponding human contact.

Ranks are computed over two prime fields. Since modular rank is a characteristic-zero lower bound and the two displayed independent exact kernel vectors give the complementary upper bound, the characteristic-zero rank is exactly 53.

This independent check is intentionally narrow. It does not replace the one-particle $290 \rightarrow 1311 \rightarrow 3657$ absolute window, full Chevalley–Eilenberg nilpotency, BV–BFV transfer or covariant causal certificates.

3 Generated-input inventory

The archival paper has one active generated TeX input. It is tracked by Git and its header binds the source certificate digest:

`generated/endpoint_factorization_nullstellensatz.tex.`

Other generated TeX files in `analytic_completion/generated/` and `covariant_completion/generated/` are tracked standalone receipts, not active `\input` dependencies of the archival paper. The future supplement manifest will classify every generated file as active input, standalone current receipt or historical checkpoint.

4 Endpoint factorization receipt

For continuity with the archival appendix, the active generated input is included below. It is a scoped no-go receipt for the complete parallel invariant two-wave-factor family; it is not used as evidence against the direct tractor causal homotopy.

$$\begin{aligned}
f_0 &= -2a_0a_4 - a_0a_5 \\
&\quad - 16a_1a_4 - 8a_1a_5 \\
&\quad - 8a_2a_4 - 4a_2a_5 \\
&\quad + 2 \\
w_0 &= \frac{3}{128} \left(18a_0^2 - 192a_0a_1 \right. \\
&\quad - 72a_0a_2 + 23a_0a_6 \\
&\quad + 384a_1^2 + 192a_1a_2 \\
&\quad + 62a_1a_6 + 58a_2a_6 \\
&\quad + a_3a_7 - 152a_4a_6 \\
&\quad + 224a_4a_8 - 68a_5a_6 \\
&\quad \left. + 112a_5a_8 \right) \\
f_1 &= 2a_0a_6 - 3a_0a_8 \\
&\quad + 16a_1a_6 - 24a_1a_8 \\
&\quad + 8a_2a_6 - 12a_2a_8 \\
&\quad - 2 \\
w_1 &= \frac{3}{128} \left(27a_0a_4 + 24a_0a_5 \right. \\
&\quad - 48a_1a_4 - 60a_1a_5 \\
&\quad + 2a_3^2 + 2a_3a_7 \\
&\quad - 224a_4^2 - 224a_4a_5 \\
&\quad - 56a_5^2 + 6a_7^2 \\
&\quad \left. - 58 \right) \\
f_2 &= -a_0a_8 + 6a_1a_6 \\
&\quad - 8a_1a_8 + 2a_2a_6 \\
&\quad - 4a_2a_8 - 2
\end{aligned}$$

$$w_2 = -\frac{3}{128} \left(81a_0a_4 + 72a_0a_5 \right. \\
- 144a_1a_4 - 180a_1a_5 \\
+ 6a_3^2 + 6a_3a_7 \\
- 224a_4^2 - 224a_4a_5 \\
- 56a_5^2 + 18a_7^2 \\
\left. - 174 \right)$$

$$f_4 = -2a_0a_3 \\
w_4 = \frac{3}{128} \left(2a_3a_6 - a_4a_7 \right. \\
\left. + a_5a_7 - 4a_6a_7 \right)$$

$$f_7 = -\frac{1}{3} \left(3a_0a_5 + 18a_1a_4 \right. \\
+ 6a_1a_5 + 6a_2a_4 \\
\left. + 6a_2a_5 - 10 \right)$$

$$w_7 = \frac{3}{128} \left(21a_0a_6 - 72a_1a_6 \right. \\
- 3a_3a_7 - 64a_4a_6 \\
\left. - 32a_5a_6 \right)$$

$$f_9 = 2a_7(a_1 - a_2) \\
w_9 = -\frac{3}{128} \left(a_3a_4 - a_3a_5 \right. \\
\left. - 6a_6a_7 \right)$$

$$f_{14} = -48a_1^2 - 48a_1a_2 \\
- 12a_2^2 + 4a_4a_6 \\
- 6a_4a_8 + 2a_5a_6 \\
- 3a_5a_8 - 1 \\
w_{14} = -\frac{1}{64} \left(3a_0^2 - 288a_0a_1 \right. \\
- 108a_0a_2 - 135a_0a_4 \\
- 36a_0a_5 - 174a_1^2 \\
- 516a_1a_2 + 432a_1a_4 \\
+ 108a_1a_5 - 174a_2^2 \\
\left. - 110 \right)$$

$$f_{15} = -16a_1^2 - 16a_1a_2 \\
- 4a_2^2 - 2a_4a_8 \\
- a_5a_8 - 1$$

$$w_{15} = \frac{9}{64} \left(a_0^2 - 96a_0a_1 \right. \\
- 36a_0a_2 - 21a_0a_4 \\
- 58a_1^2 - 172a_1a_2 \\
+ 80a_1a_4 + 4a_1a_5 \\
\left. - 58a_2^2 - 26 \right)$$

$$f_{17} = -a_0^2 + 6a_4a_6 \\
- 8a_4a_8 + 2a_5a_6 \\
- 4a_5a_8 + 4$$

$$w_{17} = -\frac{3}{64} \left(18a_0a_4 + 9a_0a_5 \right. \\
- 48a_1a_4 - 24a_1a_5 \\
\left. - 20 \right)$$

$$\begin{aligned}
f_{18} &= -\frac{1}{2} \left(2a_0^2 - 2a_3^2 \right. \\
&\quad \left. - a_4a_6 + a_5a_6 \right. \\
&\quad \left. - 4 \right) \\
w_{18} &= -\frac{3}{16} \\
f_{19} &= -a_6(a_3 + a_7) \\
w_{19} &= \frac{3}{32} \left(3a_0a_7 - a_1a_3 \right. \\
&\quad \left. + a_2a_3 \right) \\
f_{25} &= -\frac{1}{2} \left(2a_1^2 - 4a_1a_2 \right. \\
&\quad \left. + 2a_2^2 - a_4a_6 \right. \\
&\quad \left. + a_5a_6 - 2a_7^2 \right. \\
&\quad \left. + 4 \right) \\
w_{25} &= -\frac{9}{16}
\end{aligned}$$

With all omitted multipliers equal to zero, direct expansion gives

$$\sum_{i \in \{0,1,2,4,7,9,14,15,17,18,19,25\}} w_i f_i = 1.$$

The multiplier degrees are at most two. Thus the residual order-two ideal is the unit ideal over $\mathbb{Q}[a_0, \dots, a_8]$.

5 Material to extract from the archival monolith

Block	Supplement content
Finite-jet convention audit	Coordinate bases, globalization coverage and independent convention checks.
Residual ranks	Sparse formats, good-prime embeddings, pivot/minor witnesses, exact kernel representatives and the independent rank-53 check.
BV–BFV dictionaries	Raw/minimal/gauge-fixed basis ledgers, zero-mode sections, contraction maps and sign conventions.
Curvature prolongation	Weyl–Cotton row tables, sourced subsidiary system, mapping-cylinder rows, PBW normal forms and cyclic projectors.
Alternative witnesses	Scoped scalar-wave, first-order, factorization and relative-saddle searches, including exact negative receipts.
Publication provenance	Certificate schemas, content hashes, terminal proof ledger, commands, timeouts and artifact policy.

6 Submission gates

Before this scaffold becomes the submitted supplement it must:

1. build twice without undefined references;
2. have its own dependency manifest and focused verification runner;

3. reproduce every active generated input from its authoritative source;
4. distinguish theorem dependencies from historical diagnostics;
5. bind Paper A's residual claims and Paper B's causal claims to separate short ledgers;
6. preserve the archival equation-label map for referee auditability.